Ruei-Che Chang
https://rueichechang.github.io/

Toolkits for Medical Workers to Process Biopsy Image

Introduction

This work was my final project in the image processing course. I aimed to develop a simple GUI for people to deal with large and complex biopsy image. The main algorithm is Watershed Segmentation programming in C/C++ and OpenCV on the Microsoft Visual Studio. I hope this work can practically help the medial workers in the hospital, so I designed many useful functions for them to process biopsy pictures with high efficiency, including enlarging the specific part of picture, save the cell information, etc. Additionally, I designed the visual feedback for checking whether they select the target cell in the screen.

Watershed Segmentation Algorithm

Before implementing Watershed Segmentation, we usually preprocess the input image such as smoothing so as to reduce the noisy. Afterward, we do the gradient to find the minimal point as the marker, like the Figrue1, and pour the first amount of water into each catchment basin. By repeating this step until the water in two different basins encounter each other, then we build the highest dam (grayscale 255) to this match point. And the dam is the white frame showed on the above picture.

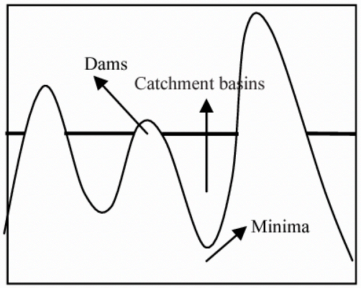
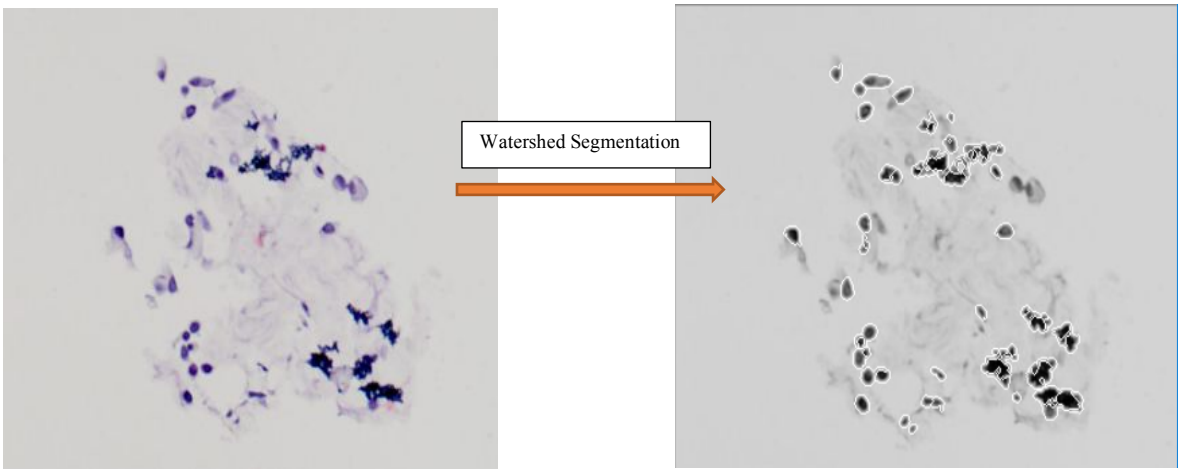


Figure1



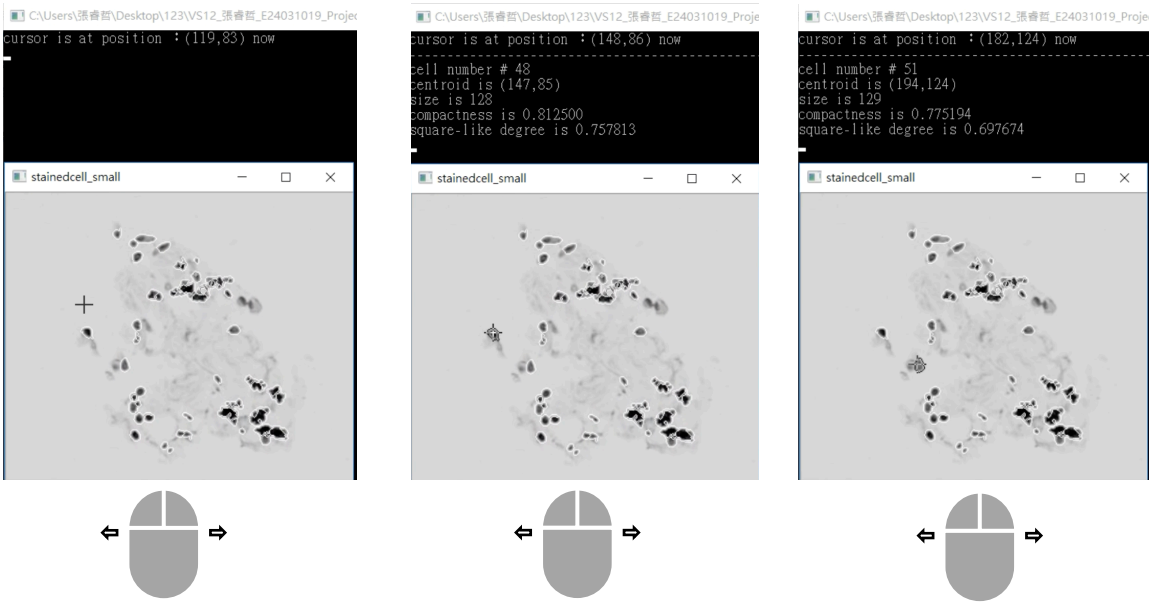
(a) (b)

Figure 2. (a) The original biopsy image. (b) The biopsy image after Watershed Segmentation.

The Figure 2 shows the original input image and the output image after processing respectively. We can see that we effectively distinguish every cell from the biopsy image. Why is this GUI important? Because the medical workers need to record the information of every cell manually, which consume much time and energy for one person. Therefore, the user interface aims to help them to efficaciously tackle with such burden.

GUI Toolkits - Cursor Moving

We design many helpful and intuitive functions for the users by only mouse click. First, in the Figure3(a), we can see the cursor moving on the output image with the information of coordinates showing on the monitor. When the cursor encounters the cell segmented by the white frame, the information of the cell instantly shows on the monitor; in the Figure3 (b) and (c), we can see the number, centroid, size and compactness of the cell. Additionally, due to the weak visibility on such small and complex image, the white frame will change into the black frame when the cursor moves on the target cell. This visual feedback is so important that many people can confirm that they find the cell.

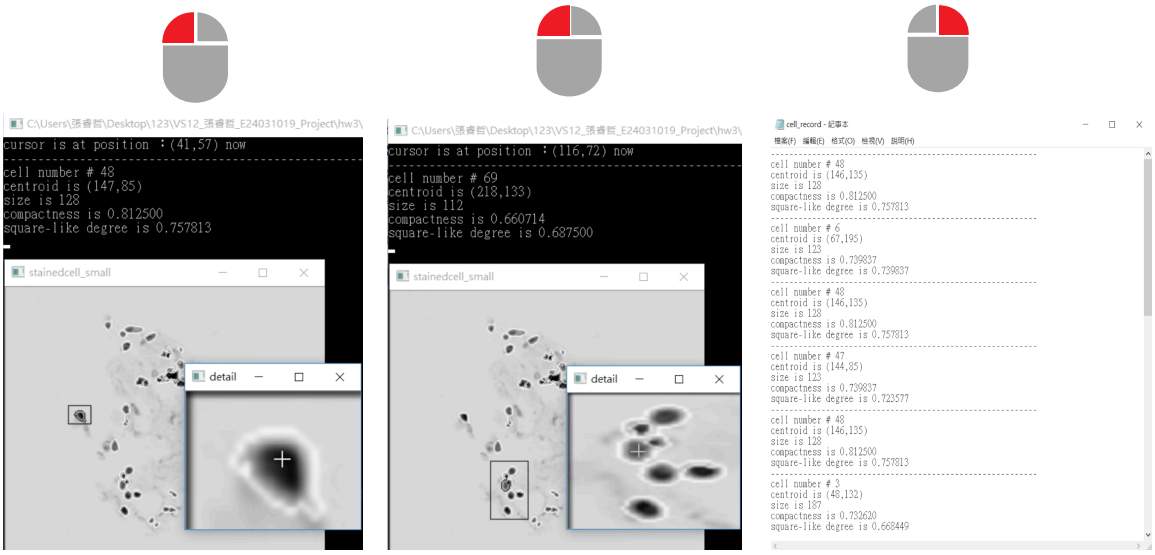


(a) (b) (c)

Figrue3. (a) when moving the cursor, the system will present the current location of which to users. (b) when moving on the specific cell, the interface will present the detailed information. (c) As the same as picture (b), but the different cell. From picture (b) and (c), we can see the visual feedback that turn the white frame to black when pointing on the cells.

Mouse Click

The mouse click is very intuitive way for human beings when manipulating the GUI. For example, the left click of mouse usually refers to select the object or magnify the target by holding the click and framing up. In the Figure4 (a), we can see that user can magnify the target cell by simply the left mouse click and the magnified cell will show up larger in the new window; simultaneously, user can also obtain the information by moving the cursor to the large cell at the new window. The Figure4 (b) shows the same thing but with the cluster of cells. When encountering the complex arrangement of cells, user can magnify the selected area and watch the information of every magnified cell. In the Figure4 (c), this is the result of the function of right mouse click. We can see the information of each cell collected in the document. This function aims to increase the efficiency of recording every features of cell by only the right mouse click. By these simple functions, the workers can deal with so many data with high efficiency.



(a) (b) (c)

Figure4. (a) our interface provides the magnify function for users to process the tiny cells. (b) users could also choose the cluster of cells and process them at the same time. (c) most medical workers rely on recording the cell information manually, which waste too much time. We offer them the function that recording the cell information by right mouse click.